

1. Levels of Organization of Life

Page 1 of 2 • Official WAEC & JAMB Syllabus Guide

Hierarchical Structure

1. Cell: Smallest basic unit of life (e.g. Amoeba, Paramecium).
2. Tissue: Group of similar cells performing a specific function (e.g. xylem, blood).
3. Organ: Group of tissues working together (e.g. leaf, heart).
4. System: Group of organs functioning together (e.g. digestive, circulatory system).
5. Organism: The complete living entity.

Cell Theory

Formulated by Schleiden and Schwann:

- All living things are composed of one or more cells.
- The cell is the structural and functional unit of life.
- All cells arise from pre-existing cells.

2. Plant vs Animal Cells & Organelles

Page 2 of 2 • Official WAEC & JAMB Syllabus Guide

Key Organelles & Their Functions

- Nucleus: Controls cellular activities and houses genetic material (DNA).
- Mitochondria: Powerhouse of the cell; site of ATP generation via respiration.
- Ribosomes: Sites of protein synthesis.
- Chloroplasts: Contain chlorophyll for photosynthesis (plants only).

Differences Between Plant and Animal Cells

- Plant Cells: Possess rigid cellulose cell wall, large central vacuole, and chloroplasts.
- Animal Cells: Lack cell walls and chloroplasts; possess centrioles and small scattered vacuoles.